Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 1 of 15 · Question 5 part b.ii · 1 mark

2023 NOV · Paper 02 · find part b.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 2 of 15 · Question 9 part b.i · 1 mark

2020 JAN · Paper 02 · find part b.i in this mark scheme extract

Number				
9 (a)	(i)	Any three from: • amino group removed/broken down(1) • in liver(1) • by deamination(1) • to produce urea (1)	Allow description of amino group e.g. nitrogenous group	3
	(ii)	Any three from: • less protein broken down by protease(1) • to amino acids 1) • in stomach/small intestine(1) • reference to reduced amount of phenylalanine/amino acids (in low protein diet)(1)		3
	(iii)	• reduced number of amino acids available(1) • during translation(1) • on ribosomes(1) • incorrect/less protein formed(1)	Allow less translation Allow less protein synthesis	3
(b)	(i)	heterozygous/Nn(1)		1
	(ii)	100% (1)		1
	(iii)	• both parents carry recessive allele/allele for PKU/heterozygous(1) • person 3 has inherited one recessive allele from each parent(1) • person 3 has PKU/homozygous recessive(1)	Reject any reference to sex-linked	3

Total for Question 9 = 14 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 3 of 15 · Question 9 part b.ii · 1 mark

2020 JAN · Paper 02 · find part b.ii in this mark scheme extract

Number				
9 (a)	(i)	Any three from: • amino group removed/broken down(1) • in liver(1) • by deamination(1) • to produce urea (1)	Allow description of amino group e.g. nitrogenous group	3
	(ii)	Any three from: • less protein broken down by protease(1) • to amino acids 1) • in stomach/small intestine(1) • reference to reduced amount of phenylalanine/amino acids (in low protein diet)(1)		3
	(iii)	• reduced number of amino acids available(1) • during translation(1) • on ribosomes(1) • incorrect/less protein formed(1)	Allow less translation Allow less protein synthesis	3
(b)	(i)	heterozygous/Nn(1)		1
	(ii)	100% (1)		1
	(iii)	• both parents carry recessive allele/allele for PKU/heterozygous(1) • person 3 has inherited one recessive allele from each parent(1) • person 3 has PKU/homozygous recessive(1)	Reject any reference to sex-linked	3

Total for Question 9 = 14 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 4 of 15 · Question 9 part b.iii · 3 marks

2020 JAN · Paper 02 · find part b.iii in this mark scheme extract

Number				
9 (a)	(i)	Any three from: • amino group removed/broken down(1) • in liver(1) • by deamination(1) • to produce urea (1)	Allow description of amino group e.g. nitrogenous group	3
	(ii)	Any three from: • less protein broken down by protease(1) • to amino acids 1) • in stomach/small intestine(1) • reference to reduced amount of phenylalanine/amino acids (in low protein diet)(1)		3
	(iii)	• reduced number of amino acids available(1) • during translation(1) • on ribosomes(1) • incorrect/less protein formed(1)	Allow less translation Allow less protein synthesis	3
(b)	(i)	heterozygous/Nn(1)		1
	(ii)	100% (1)		1
	(iii)	• both parents carry recessive allele/allele for PKU/heterozygous(1) • person 3 has inherited one recessive allele from each parent(1) • person 3 has PKU/homozygous recessive(1)	Reject any reference to sex-linked	3

Total for Question 9 = 14 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 5 of 15 · Question 6 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
6	(a) (i)	any three from • prevents excessive loss of blood (1) • forms scab over wound (1) • prevents entry of bacteria /pathogens/microorganisms(1) • so prevents infection (1) • allows wound to heal (1)		3
	(ii)	• (allele) on X/sex chromosome (1) • passed on to offspring/inherited by children (1)		2
	(b) (i)	D; (X^hY) A the genotype is not X^hX^h B the genotype is not X^HX^h C the genotype is not X^HY		1
	(ii)	B; (X^HX^h) A the genotype is not $X^h X^h$ C the genotype is not X^HY D the genotype is not X^hY		1
	(iii)	• offspring genotypes X^HX^h and X^hY (1) • 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1)	full marks for correct figures only.	2
	(iv)	• unusual for female to be haemophiliac (1) • both alleles have to be affected (1) • must have a haemophiliac father and mother with haemophilia/carrier (1)	any correct reference to excessive blood loss	3

Total for question = 12 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 5 of 15 · Question 6 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 6 of 15 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
6	(a) (i)	any three from • prevents excessive loss of blood (1) • forms scab over wound (1) • prevents entry of bacteria /pathogens/microorganisms(1) • so prevents infection (1) • allows wound to heal (1)		3
	(ii)	• (allele) on X/sex chromosome (1) • passed on to offspring/inherited by children (1)		2
	(b) (i)	D; (X^hY) A the genotype is not X^hX^h B the genotype is not X^HX^h C the genotype is not X^HY		1
	(ii)	B; (X^HX^h) A the genotype is not $X^h X^h$ C the genotype is not X^HY D the genotype is not X^hY		1
	(iii)	• offspring genotypes X^HX^h and X^hY (1) • 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1)	full marks for correct figures only.	2
	(iv)	• unusual for female to be haemophiliac (1) • both alleles have to be affected (1) • must have a haemophiliac father and mother with haemophilia/carrier (1)	any correct reference to excessive blood loss	3

Total for question = 12 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 6 of 15 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 7 of 15 · Question 6 part b.i · 5 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

6	(a) (i)	• prevent excess blood loss; • prevent entry of pathogens/bacteria/microorganisms;		2
	(ii)	V = platelets; W = thrombokinase enzyme; Y = fibrinogen; Z = fibrin;		4
	(b) (i)	• carried on X chromosome; • male must be X^hY; • female must be X^HX^h; • as allele is recessive; • but appears in next generation; any four from		5
	(ii)	• female has X^hX^h genotype; • so male offspring must be X^hY; • probability of a boy is 0.5; • probability of haemophilia is 1.0; • so probability of a boy with haemophilia is 1.0 × 0.5 = 0.5		4

Total 15 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 7 of 15 · Question 6 part b.i · 5 marks

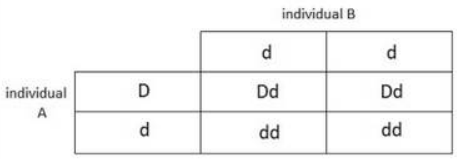
2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 8 of 15 · Question 7 part c.i · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

Number				
7	(a)	Any three from: • changes the DNA/base/codon sequence; • changes the order of the amino acids/different amino acid coded for; • changes the function/shape of the protein/different protein; • affects a characteristic or function/genotype in the individual;		3
	(b) (i)	Two from: • dialysis has to be completed several times a week/transplant permanent; • increased life expectancy; • less restrictions on diet; • better quality of life/freedom to move around/normal lifestyle;		2
	(ii)	100 x 3233÷4733; = 68.3/68		2
	(iii)	Two from: • people have two kidneys; • two recipients from one donor; • living donation/can live with just one;		2
	(c) (i)	0.5:1;	Accept 1:2	1
	(ii)	 • correct genotype individual A; • correct genotype individual B; • 50% offspring heterozygous/correct genotype and affected; • 50% homozygous recessive/correct genotype and not affected;		4

Total 14 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 8 of 15 · Question 7 part c.i · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 9 of 15 · Question 6 part b · 4 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number				
6	(a) (i)	• change in genetic code/base sequence/nucleotide sequence/different codon; • of DNA/gene; • different amino acid coded for/different sequence of amino acids; • different (shaped) protein produced;	Allow named mutation e.g. deletion	4
	(ii)	Any five from • shape of red blood cell distorted; • different form of haemoglobin; • reduced surface area; • less oxygen carried; • less respiration; • less energy/ATP;	Allow red blood cells are sickle-shaped Allow capillaries blocked/allow less haemoglobin	5
	(b)	• woman is Hh; • man is either Hh or hh/heterozygous or homozygous recessive; • one h/recessive allele comes from man; • child is hh/homozygous recessive;		4

Total 13

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 9 of 15 · Question 6 part b · 4 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 10 of 15 · Question 5 part b.i · 2 marks

2023 NOV · Paper 02 · find part b.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 11 of 15 · Question 4 part c.i · 2 marks

2024 JUN · Paper 02 (regional R) · find part c.i in this mark scheme extract

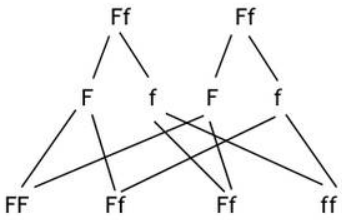
4 (a)	(i)	11, 31, 30		(1)
	(ii)	31 (× 100); 100 = 31%;	send to review if other than 31	(2)
(b)	(i)	**D (alternative forms of the same gene in a genotype)** The following are incorrect because: A two dominant genes are the same type of allele B two recessive genes are the same type of allele C alleles are not different genes		(1)
	(ii)	• sperm and egg/gametes each contain one allele/haploid; • sperm and egg/gametes fuse; • during fertilisation; • (fertilisation) brings alleles together (in diploid cell)/chromosomes (from each gamete) are paired/one allele comes from each parent;		(4)
(c)	(i)	Hh/heterozygous; hh/homozygous recessive;	either order any suitable letters to be used for alleles	(2)
	(ii)	wheezing/coughing/shortness of breath/low weight / fatigue/over production of mucus/diarrhoea/male infertility;	allow other valid symptoms	(1)

Total for Question 4 = 11 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 12 of 15 · Question 8 part a.i · 1 mark

2024 JUN · Paper 02 · find part a.i in this mark scheme extract

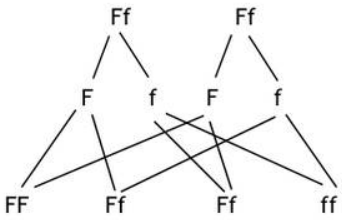
Number				
8	(a) (i)	**B** individual 3 The following are incorrect because: **A** individual 1 is heterozygous **C** Individual 5 is heterozygous **D** individual 3 is homozygous dominant		(1)
	(ii)	**D** 4 The following are incorrect because: **A** there are more than 1 heterozygous genotypes **B** there are more than 2 heterozygous genotypes **C** there are more than 3 heterozygous genotypes		(1)
	(iii)	• they have a dominant allele; • over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ;	do not allow dominant allele overpowering the recessive allele	(2)
	(iv)	• heterozygous parent genotypes; • gametes; • genotypes of offspring; • offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;  	mps 2,3 and 4 marks for correct Punnett square showing all details	(4)
	(b)	• reference to use of restriction enzyme • (to obtain) the normal/ non faulty/ dominant allele); • use of virus/ correctly named virus; • inserts allele into person **cells**; • functional/normal protein produced;	Accept liposomes ignore plasmids Penalise one for using genes instead of alleles.	(3)

Total for Question 8 = 11 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 13 of 15 · Question 8 part a.ii · 1 mark

2024 JUN · Paper 02 · find part a.ii in this mark scheme extract

Number				
8 (a)	(i)	**B** individual 3 The following are incorrect because: **A** individual 1 is heterozygous **C** Individual 5 is heterozygous **D** individual 3 is homozygous dominant		(1)
	(ii)	**D** 4 The following are incorrect because: **A** there are more than 1 heterozygous genotypes **B** there are more than 2 heterozygous genotypes **C** there are more than 3 heterozygous genotypes		(1)
	(iii)	• they have a dominant allele; • over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ;	do not allow dominant allele overpowering the recessive allele	(2)
	(iv)	• heterozygous parent genotypes; • gametes; • genotypes of offspring; • offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;  	mps 2,3 and 4 marks for correct Punnett square showing all details	(4)
	(b)	• reference to use of restriction enzyme • (to obtain) the normal/ non faulty/ dominant allele); • use of virus/ correctly named virus; • inserts allele into person **cells**; • functional/normal protein produced;	Accept liposomes ignore plasmids Penalise one for using genes instead of alleles.	(3)

Total for Question 8 = 11 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 14 of 15 · Question 7 part a.i · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

7 (a)	(i)	• 2nd set of parents have genotypes I^BI^O and I^BI^O/both parents are I^BI^O; • give rise to child I^OI^O which is M; • 1st set of parents have genotypes I^OI^O and I^AI^B; • can give rise to a child I^AI^O which is N;	Allow 1 mark if the child is matched to the correct parent.	4
	(ii)	• parental genotypes $I^BI^O \times I^BI^O$; • offsprings are I^BI^B 2 I^BI^O I^OI^O; • 75%/0.75/3:1/3 in 4;	accept parental gametes for mp1 if BO and BO parental genotypes are used they can get ecf for MP2. accept offsprings in genetic cross	3
	(b)	• (prick finger) with pin/needle; • sterile; • (smear) blood on slide; • thinly; • dry; • stain;	accept example of sterile Accept MP5 and MP6 in either order	5
				Total =12

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 14 of 15 · Question 7 part a.i · 4 marks

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Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 15 of 15 · Question 5 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

5 (a)	(i)	• change in nucleotide/base/codon sequence; • of DNA;		2
	(ii)	• new/different (sequence of) amino acids; • different protein / enzyme / different polypeptide • different phenotype/enzyme does not bind to substrate;		3
(b)	(i)	• allele for haemophilia on X chromosome/ not on Y chromosomes; • hairy ears could only be males / haemophilia could occur in females (and males);		2
	(ii)	• 1 = XX; • 2 = XY^H;		2
	(iii)	• chance of being a girl = 50%/ this can be indicated in a punnet square; • girl will have XX genotype / girl indicated as XX from punnet square/ girls have two X; • no Y chromosome so no chance of hairy ears;		3

Total 12 marks

Type MS · Deduce genotypes from a pedigree, family or blood group

Answer 15 of 15 · Question 5 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks